MIL-C-62074C(AT)
15 May 1985
SUPERSEDING
MIL-C-62074B(AT)
19 June 1978

MILITARY SPECIFICATION

CHASSIS, WEAPON, FULL-TRACKED, M741 AND M741A1; PROCESSING FOR STORAGE AND SHIPMENT OF

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Levels A and B processing of the Chassis, Weapon, Full-Tracked, M741 and M741A1 for storage and shipment (see 1.2).

1.2 Classification. Processing shall be of the following levels, as specified (see 6.2).

Level A - Maximum military protection.

Level A is the processing required for the protection of vehicle during shipment, handling, and storage exceeding 90 days from date of actual processing. This level does not provide for driveaway capability. It does provide for domestic or overseas shipment, including open deck loading.

Level B - Minimum military protection.

Level B is the limited processing required for the protection of vehicle during shipment, handling, and storage not to exceed 90 days from date of actual processing. This level provides for driveaway capability, when specified, and domestic or overseas shipment (excluding open deck loading).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GSS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AREA PACK

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS
FEDERAL

O-I-490	- Inhibitor, Corrosion, Liquid Cooling System.
O-S-576	- Sodium Bicarbonate, Technical.
O-S-801	- Sulfuric Acid, Electrolite; for Storage Batteries.
P-D-220	- Detergent, General Purpose.
FF-N-105	- Nails, Brads, Staples and Spikes: Wire, Cut and Wrought.
GG-T-250	- Tester, Antifreeze Solutions.
NN-P-530	- Plywood, Flat Panel.
QQ-S-781	- Strapping, Steel, and Seals.
QQ-W-470	- Wire, Steel, Carbon, Spring, Music.
RR-W-360	- Wire Fabric, Industrial.
TT-E-529	- Enamel, Alkyd, Semi-Gloss.
TT-I-1795	- Ink, Marking, Stencil, Opaque (Porous and Nonporous Surfaces).
TT-V-121	- Varnish, Spar, Water-Resisting.
UU-P-268	- Paper, Kraft, Wrapping.
UU-T-81	- Tags, Shipping and Stock.
VV-F-800	- Fuel Oil, Diesel.
VV-L-800	- Lubricating Oil, General Purpose, Preservative (Water-Displacing, Low Temperature).
ZZ-T-416	- Tire, Pneumatic, Retread and Repair Materials.
MMM-A-178	- Adhesive, Paper Label, Water-Resistant.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-621	- Boxes, Wood, Nailed, and Lock-Corner.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-C-1752	- Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible.
PPP-T-42	- Tape, Packaging/Masking, Paper.
PPP-T-60	- Tape, Packaging, Waterproof.
PPP-T-97	- Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

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MIL-P-116	- Preservation, Methods of.
MIL-B-117	- Bag, Sleeve and Tubing-Interior Packaging.
MIL-C-450	- Coating Compound, Bituminous Solvent Type, Black (for Ammunition).
MIL-C-5501	- Cap and Plug, Protective, Dust and Moisture Seal.
MIL-H-6083	- Hydraulic Fluid, Petroleum Base, for Preservation and Operation.
MIL-B-11188	- Battery, Storage, Lead Acid.
MIL-A-11755	- Antifreeze, Arctic-Type.
MIL-B-12841	- Basic Issue Items for Military Vehicles, Carriages and Equipment; Preparation for Shipment and Storage of.
MIL-P-13983	- Paint, Temporary, Lusterless, Gasoline Removable.
MIL-P-14232	- Parts, Equipment and Tools for Army Material, Packaging and Packing of.
MIL-C-16173	- Corrosion Preventive Compound, Solvent Cutback, Cold Application.
MIL-D-16791	- Detergent, General Purpose (Liquid, Nonionic).
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-T-22085	- Tape, Adhesive, Preservation and Sealing.
MIL-B-22191	- Barrier Materials, Transparent, Flexible, Heat Sealable.
MIL-P-46002	- Preservative Oil, Contact and Volatile Corrosion Inhibited.
MIL-A-46153	- Antifreeze, Ethylene Glycol Inhibited, Heavy Duty, Single Package.
MIL-T-50036	- Talc, Technical, Tl and T3.
MIL-H-52079	- Hose Preformed, Semiflexible, Reinforced.
MIL-D-81298	- Dye, Liquid Red, for the Detection of Leaks in Aircraft Fuel Systems.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-731	- Quality of Wood Members for Containers and Pallets.
MS 27183	- Washer, Flat-Round, Steel, Cadmium Plated, General Purpose (in/mm).
MS 35842	- Clamp Hose: Low Pressure, Type "F".
MS 51922	- Nut, Self-Locking, Hexagon - Prevailing Torque, General Purpose, 250°F, UNC-2B and UNF-2B.
MS 90725	- Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Grade 5, Cadmium Plated, UNC-2A.

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASSOCIATION OF AMERICAN RAILROADS PUBLICATIONS

Section No. 1	- General Rules Governing Loading of Commodities on Open Top Cars.
Section No. 6	- Rules Governing the Loading of Defense Material on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren, Chicago, Illinois 60605.)

DEPARTMENT OF TRANSPORTATION (DOT)

Hazardous Materials Regulations

(Application for copies should be addressed to the Department of Transportation, Hazardous Materials Regulations Board, Washington, DC 20590.)

Federal Motor Carrier Safety Regulations

(Application for copies should be addressed to the Department of Transportation, Bureau of Motor Carriers, Washington, DC 25090.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), one of the first 10 production processed vehicles shall be subjected to the inspection specified in 4.2. Approval of this vehicle shall not relieve the contractor of his obligation to process all vehicles in accordance with this specification. Unless otherwise specified by the Government, any change to materials or design after approval shall require additional vehicles to be inspected as specified in 4.2. The Government representative shall select the vehicle to be inspected.

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3.2 Materials. Materials shall be as specified herein and in referenced specifications and standards. Materials shall be free of defects which adversely affect performance or serviceability (see 4.5.1 and 6.5).

3.3 Processing levels. Level A processing shall be as specified in 3.4. Level B processing shall be as specified in 3.5.

3.4 Level A.

3.4.1 Preparation prior to processing. Except as otherwise specified herein, and to the maximum extent consistent with production efficiency, economy, and safe storage and shipment, the vehicle shall be prepared for storage and shipment in a completely assembled condition after test runs and completion and approval of all necessary repairs. Specified equipment shall be installed and all adjustments made so that the vehicle may be operated, shipped, and placed into service with a minimum of delay.

3.4.1.1 Record forms. Two copies of DD Form 1397 shall be provided. Information on forms shall include preservation accomplished and depreservation instructions. The Equipment Log Book Binder and one copy of DD Form 1397 shall be placed in a bag conforming to type I, class B, style 2 of MIL-B-117. The bag shall be closed by heat sealing and securely attached in driver's compartment of the vehicle. The other copy of DD Form 1397 shall be waterproofed with adhesive conforming to MMM-A-178 or sealed in a transparent plastic bag and securely attached in a conspicuous location on the exterior of the vehicle.

3.4.1.2 Processing records. Records of vehicle processing shall be maintained and shall be readily available for review by the Government representative.

3.4.2 Government furnished equipment. Unless previously accomplished, Government furnished equipment (GFE) (other than installed) shall be packaged, packed and marked in accordance with the individual document for the specific item. GFE shall be stowed with basic issue items (BII).

3.4.3 Preservatives and atomized spray equipment. When atomized spraying of preservative oils is specified, equipment shown in figures 1 and 2, or equivalent, shall be used. Compressed air supply lines shall be equipped with moisture separators every 50 feet or fraction thereof.

3.4.4 Disassembly. Projecting parts whose removal will accomplish desired cube reduction and parts susceptible to damage and pilfereage shall be removed from the vehicle. Removed bolts, nuts, screws, pins, and washers shall be placed in one of the mating parts and secured. Removed parts shall be preserved, packaged, and packed. Packed parts shall be identified and stowed securely within the vehicle.

3.4.4.1 Matchmarking. Parts removed from the vehicle shall be matchmarked when necessary to facilitate reassembly. Matchmarking information shall be put on cloth shipping tags conforming to type A of UU-T-81, or on metal tags using waterproof ink or paint, and attached to mating parts. The marked cloth shipping tags shall be waterproofed with varnish conforming to TT-V-121 or adhesive conforming to MMM-A-178.

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3.4.5 Cleaning and drying.

3.4.5.1 Interior of vehicle. Interior surfaces of the vehicle shall be cleaned with a solution of detergent conforming to P-D-220 or type I of MIL-D-16791 and warm water. Water or other liquid under pressure or steam cleaning shall not be used. After cleaning, cleaned surfaces shall be rinsed with clean water and dried. Care shall be taken during cleaning and rinsing to assure that no liquids enter instruments, connections, or other components susceptible to water damage and that water does not accumulate in areas where it cannot drain or be dried (see 4.5.2).

3.4.5.1.1 Battery supports and retainers. Battery supports and retainers shall be cleaned with a solution of one-half pound of sodium bicarbonate conforming to O-S-576 per gallon of water. After cleaning, cleaned surfaces shall be flushed with clean water, then thoroughly dried. Dried surfaces shall then be preserved in accordance with 3.4.7.

3.4.5.1.2 Backrests and seats. Backrest and seat cushions shall be cleaned with a solution of detergent conforming to P-D-220, or type I of MIL-D-16791, in warm water. After cleaning, cushions shall be wiped with cloths saturated with clean water to remove cleaning solution. Care shall be taken not to saturate cushions with cleaning solution or water. After rinsing, cushions shall be dried, then protected in accordance with 3.4.19.3.

3.4.5.2 Vehicle exterior. The vehicle exterior shall be cleaned using a solution of detergent conforming to P-D-220, or type I of MIL-D-16791, in warm water or steam. Cleaning shall remove all foreign matter. After cleaning, cleaned surfaces shall be rinsed with clean water or steam and thoroughly dried. Care shall be taken to avoid entry of water or steam into driver's or engine compartments (see 4.5.2).

3.4.6 Relubrication. If the vehicle has been operated more than 75 miles since lubrication, or after the vehicle has been cleaned in accordance with 3.4.5.2, it shall be relubricated using materials conforming to applicable drawings, specifications, or lubrication orders. All exposed oil can points such as, but not limited to, levers, locking levers, locking bars, locking pins, pintle pins, hing pins, hinges, strikers, wing nuts, door locks, hand-operated locking knobs, latches, linkage, and threaded ends of yokes and related clevis pins shall be coated with lubricant conforming to VV-L-800. Excess lubricant shall be removed after coating.

3.4.7 Preservation of battery supports and retainers. Top battery supports and retainers shall be preserved with compound conforming to MIL-C-450.

3.4.8 Transmission, transfer assembly, control differential and final drives. The transmission shall contain lubricating oil conforming to grade PE 10-1 of MIL-L-21260 filled to the operating level. The transfer assembly, control differential, and final drives shall contain lubricating oil conforming to grades PE 10-1 or PE 30-1, as applicable, of MIL-L-21260 filled to the operating level. DD Form 1397 shall be annotated with the grade of lubricant used (see 3.4.1.1).

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3.4.9 Cooling system. The cooling system shall be protected by one of the following procedures, as specified (see 4.5.3 and 6.2):

a. For shipment to, and storage in, areas where the temperature drops below minus 40 degrees Fahrenheit (°F), systems shall be protected as specified in 3.4.9.3.
b. For shipment and storage within the bounds of 30 degrees north latitude and 20 degrees south latitude, except continental United States, systems shall be protected as specified in 3.4.9.2.
c. For all other shipments, cooling systems shall be protected as specified in 3.4.9.1.

NOTE: DD Form 1397 (see 3.4.1.1) shall be completed to indicate coolant used.

3.4.9.1 Water and antifreeze procedure. The cooling system shall be filled to the operating level with a clean solution consisting of equal parts by volume of antifreeze (ethylene glycol) conforming to MIL-A-46153, and water. The engine shall be operated until a temperature has been reached that causes the thermostat to open, to assure complete mixing and even distribution of the antifreeze solution. A warning tag, bearing the information "COOLING SYSTEM FILLED WITH WATER AND ANTIFREEZE SOLUTION (ETHYLENE GLYCOL) IN EQUAL PARTS BY VOLUME - DO NOT DRAIN", shall be securely attached to the radiator filler neck.

3.4.9.2 Water and corrosion inhibitor procedure. The cooling system shall be filled with clear water up to, but not including the radiator upper tank. A corrosion inhibitor conforming to O-I-490 shall be added in the proportion of five ounces of the inhibitor for each ten quarts of water. The inhibitor shall be dissolved in two quarts of warm water and poured into the radiator while the engine is idling. More water shall be added, if necessary, to fill the radiator to operating level. A warning tag, bearing the information "COOLING SYSTEM DOES NOT CONTAIN ANTIFREEZE - FILLED WITH WATER AND INHIBITOR", shall be securely attached to the radiator filler neck.

3.4.9.3 Antifreeze compound procedure. The cooling system shall be filled to the operating level with antifreeze compound conforming to MIL-A-11755. The compound shall be used without dilution. A warning tag, bearing the information "COOLING SYSTEM FILLED WITH ANTIFREEZE (ARCTIC-TYPE) - DO NOT DRAIN", shall be securely attached to the radiator filler neck.

3.4.10 Engine crankcase preservation. The engine crankcase shall be filled to the operating level with lubricating oil conforming to MIL-L-21260 of the seasonal grade specified in the applicable drawing, specification, or lubrication order. DD Form 1397 shall be annotated with the type and grade of lubricant used (see 3.4.1.1).

3.4.11 Engine preservation. Engine preservation shall be in accordance with 3.4.11.1 through 3.4.11.3 in an uninterrupted sequence.

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3.4.11.1 Fuel system and combustion chamber. Prior to processing, the engine shall be cooled to assure that the cylinder head temperature, measured at the injector nozzle flange surface of all cylinders, is not more than 100°F. Cooling shall be accomplished by induced air currents, circulation of engine coolant, or by waiting the period of time required to attain the above specified temperature. When the ambient temperature exceeds 100°F, the engine shall be cooled to ambient temperature. After the engine has been cooled, the fuel supply system from the fuel tank shall be shut off. A portable container with two compartments shall be positioned to provide gravity feed to the engine. One compartment shall be filled with preservative oil conforming to grade 1 of MIL-P-46002 to which has been added an oil soluble red dye conforming to MIL-D-81298, in a concentration sufficient to impart a marked coloring to the oil. The second compartment shall be filled with diesel fuel conforming to VV-F-800. (The container shown in figure 3 has proven satisfactory for engine preservation.) Disconnect the fuel line between the primary fuel filter and fuel pump at the primary fuel filter outlet. Drain the diesel fuel from the secondary fuel filter. Remove the filter can and remove the filter element. Fill the filter can with preservative oil, conforming to grade 1, MIL-P-46002 and reinstall it. Connect the fuel line from the auxiliary fuel container to the fuel pump supply line. Disconnect the vehicle fuel return line at the quick disconnect coupling. Connect a transparent plastic fuel line to the engine end of the disconnected fuel return line. Insert the other end of the plastic fuel line into a recovery container to collect the returned oil. Disconnect the air cleaner hose from the air inlet elbow. An air restrictor boot shall be installed on the air inlet elbow and secured with a clamp to assure that the air to the engine is completely cut off. (The air restrictor boot shown in figure 4 has proven satisfactory for engine preservation.) The regulator valve on the auxiliary fuel ocntainer shall be turned to the preservative oil position. Place the engine fuel control in run position. Crank the engine with the starter for a period of not less than 30 seocnds and not more than 45 seconds (NOTE: The engine may fire for approximately five seconds). The regulator valve on the auxiliary fuel container shall be turned to the off position. Remove the filter can from the secondary fuel filter. Drain the preservative oil, grade 1, MIL-P-46002, from the can and wipe it clean with lint free cloth. Place the filter element in the filter can and fill the can with diesel fuel VV-F-800. Reinstall the filter can in the original position (see 4.5.4).

CAUTION: Special precautions shall be taken to assure that time limits specified shall not be exceeded, or the engine starter or starter solenoid may be damaged. Excessive cranking may also cause hydrostatic locks by injecting too much oil into combustion chambers.

3.4.11.2 Engine purging. Purge fuel return lines of all preservative oil, by turning the auxiliary fuel tank valve to the diesel fuel position and closing the engine fuel control shut off. Crank the engine with the starter to remove all traces of red colored lubricating oil from the plastic return line. CAUTION: DO NOT OPERATE STARTER FOR A PERIOD EXCEEDING 45 SECONDS. If all preservative oil has not been purged, rest the starter for three minutes and repeat the purge procedure. Turn the auxiliary fuel container

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valve to the OFF position. Remove the fuel line and reconnect it to the outlet side of the primary fuel filter. Turn on the vehicle fuel supply system. Remove the air restrictor boot. Disconnect the plastic fuel return line, and reconnect vehicle fuel return line at the quick disconnect coupling (see 4.5.4).

3.4.11.3 Preservation through air intake and exhaust system. After preservation in accordance with 3.4.11.1 and 3.4.11.2, one ounce of preservative oil conforming to grade 1 of MIL-P-46002 shall be atomized into the exhaust opening. The opening shall then be sealed with tape conforming to type IV, class 1 of PPP-T-60 or type II of MIL-T-22085. The hose shall be disconnected at the air intake, and one ounce of preservative oil, conforming to grade 1, MIL-P-46002, shall be atomized into the air intake opening. The air intake opening and the engine crankcase breathers shall be sealed with plastic plugs conforming to MIL-C-5501 or with tape conforming to type IV, class 1 of PPP-T-60 or type II of MIL-T-22085. A red warning tag, bearing the information "ENGINE PRESERVED WITH VCI - DO NOT CRANK" and "BEFORE CRANKING, REMOVE CAPS, TAPE, OR PLUGS, AND REINSTALL HOSE", shall be placed in a conspicuous location within driver's compartment. DD Form 1397 shall be annotated to show engine is preserved with volatile corrosion-inhibited (VCI) oil (see 3.4.1.1 and 4.5.4).

3.4.11.4 Preservation through oil level gage rod opening. After preservation in accordance with 3.4.11.1 and 3.4.11.2, the oil level gage rod shall be removed, and six ounces of preservative oil conforming to grade 1 of MIL-P-46002 shall be atomized into the crankcase through the oil level gage rod opening. An extension of sufficient length to permit the spray nozzle to be within the crankcase shall be used. The spray nozzle shall not be submerged in the crankcase oil. After spraying has been accomplished, the oil level gage rod shall be reinstalled. All openings to the engine interior, including the oil level gage rod opening and the oil filler cap, shall be sealed with tape conforming to type IV, class 1 of PPP-T-60 or type II of MIL-T-22085 (see 4.5.4).

3.4.11.5 Preservation through flywheel housing. Two ounces of preservative oil conforming to grade 1 of MIL-P-46002 shall be atomized into the flywheel housing.

3.4.11.6 Air cleaner. Air cleaner openings shall be sealed with tape conforming to type IV, class 1 of PPP-T-60 or type II of MIL-T-22085.

3.4.12 Personnel and engine compartment heaters and lines. The personnel and engine heaters shall have the fuel supply shut off valve located at the inlet side of the fuel filters, turned to the OFF position. The main fuel line supplying fuel to the heaters shall be disconnected at a point closest to the shut off valves. Fuel from the fuel lines shall be allowed to drain. Seal the ends of the disconnected fuel lines and the shut off valves with plastic plugs/caps conforming to MIL-C-5501, or with tape conforming to type IV, class 1 of PPP-T-60, or with type II of MIL-T-22085. External exhaust stack shall have opening sealed with tape conforming to type IV, class 1 of PPP-T-60 or type II of MIL-T-22085. Plastic plug/cap

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conforming ot MIL-C-5501, may be used. Four warning tags, each bearing the information "HEATER FUEL LINES DISCONNECTED AND SEALED. PRIOR TO PLACING PERSONNEL OR ENGINE HEATERS IN OPERATION, REMOVE PLUGS/CAPS OR TAPE FROM FUEL LINES, EXHAUST STACK AND SHUT OFF VALVES. OPERATE HEATER FUEL PUMP AND DRAIN A MINIMUM OF ONE QUART OF FUEL THROUGH THE FUEL LINES INTO A PORTABLE CONTAINER. RECONNECT HEATER FUEL LINES.", shall be prepared. One tag each shall be secured to the personnel and engine heater operating switches and one each to the personnel and engine heaters.

3.4.13 Fuel tank preservation. The fuel tank shall be drained to the maximum extent possible. The fuel tank cap and the filler screen shall be removed and coated with lubricating oil conforming to grade PE 30-1 of MIL-L-21260. One quart of lubricating oil conforming to grade PE 10-1 of MIL-L-21260 shall be added to each five gallons or portion thereof of residual fuel that remains after draining the tank. The contractor shall maintain a written procedure used to ascertain the amount of residual fuel. The tank cap and filler screen shall be reinstalled (see 4.5.5).

3.4.13.1 Fuel tank security. After processing the fuel tank as specified in 3.4.13, the armored fuel cap shall be secured with the combat lock.

3.4.14 Ramp winch assembly. All unpainted metal surfaces of the ramp winch assembly, excluding cylinder rod, shall be coated with preservative conforming to grade 4 of MIL-C-16173.

3.4.14.1 Ramp hydraulic reservoir. Ramp hydraulic reservoir shall be filled with hydraulic fluid conforming to MIL-H-6083.

3.4.15 Turret opening. Turret opening shall be provided with a cover. Cover shall be fabricated and installed in accordance with figures 5 and 6.

3.4.16 Hatches and doors. Rubber seals around hatches and doors shall be coated with powdered talc conforming to type IV, class C of ZZ-T-416 or conforming to technical talc conforming to MIL-T-50036. For shipment, hatches and doors shall be closed and locked from the inside. Driver's hatch shall be closed and secured with a bolt and nut with thread portion of bolt deformed or nut welded to the bolt. For storage, hatches and doors shall be locked from the inside, except that the ramp door shall be secured in open position for ventilation.

3.4.17 Ventilation.

3.4.17.1 Engine compartment access plate and drain plugs. The engine compartment access plate, gasket, and attaching hardware shall be removed for ventilation. A screen conforming to figure 7 shall be installed in access plate openings, and secured with four of the existing mounting screws and washers. The two forward MS drain plugs and the rear bilge drain plugs shall be removed for drainage. A screen conforming to figure 7A shall be installed in rear bilge opening, and held in place with retainer spring conforming to figure 7B. Bare metal surfaces of drain plugs shall be preserved with

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compound conforming to grade 4 of MIL-C-16173. Plate, gasket, preserved drain plugs, and four access plate mounting screws and washers shall be packaged as specified 3.4.19.7. The information "REMOVE SCREENS, INSTALL ACCESS PLATE AN GASKET, FRONT AND REAR DRAIN PLUGS BEFORE VEHICLE OPERATION" shall be stenciled on the exterior of the vehicle using white or yellow paint conforming to MIL-P-13983. Stenciling shall be in characters not less than 0.75 inch high.

3.4.17.2 Engine access compartment panels. Only the lower rear engine compartment panel in crew compartment shall be removed, and stowed securely in crew compartment. A warning tag, bearing the information "ENGINE COMPARTMENT PANEL REMOVED: LOWER RAMP OR OPEN HATCHES WHEN OPERATING ENGINE" shall be attached in a conspicuous location within the driver's compartment.

3.4.18 Miscellaneous preservation. Except as otherwise specified herein, all exposed, unpainted metal surfaces on the exterior of the vehicle, except the track shoes, shall be coated with compound conforming to Grade 1 of MIL-C-16173. All exposed, unpainted, unplated metal surfaces on the interior of the vehicle shall be coated with compound conforming to Grade 4 of MIL-C-16173.

3.4.19 Packaging.

3.4.19.1 Dry charged batteries and cables. Dry charged batteries shall be installed and secured in the vehicle battery carrier. Battery cables shall be secured to the battery carrier with 0.75 inch tape conforming to type IV of PPP-T-97. Battery filler cap openings shall be sealed by placing a 2 inch wide by 3 mil thick piece of film conforming to type II of MIL-B-22191 over each filler cap opening with the cap removed. The sheet shall be of sufficient length to allow it to be depressed into the opening to the same depth as the filler plug. Filler caps shall be screwed or inserted into openings to form a complete seal without damaging the sheet. If batteries have been processed in accordance with MIL-B-11188, they need not be reprocessed as above.

3.4.19.2 Electrolyte. Electrolyte shall be packaged and packed in accordance with O-S-801, paragraph 5.1.1 and 5.2.1 (as applicable), except that the exterior container shall conform to class 2 of PPP-B-621 or overseas type of PPP-B-601. Marking shall be according to paragraph 5.3 of O-S-801. The packed electrolyte shall be stowed in the same location as the BII and secured independently to permit separate removal.

3.4.19.3 Packaging of backrests and seats. Cushions of backrests and seats (see 3.4.5.1.2) shall be covered with paper conforming to grade B of UU-P-268 with a minimum basis weight of 60 pounds. Paper shall be secured with tape conforming to type I of PPP-T-42.

3.4.19.4 Periscopes. If installed, periscopes shall be removed from vehicle, cleaned, dried, and immediately packaged and packed in accordance with Level A requirements of MIL-P-116 and MIL-P-14232, then securely stwoed within personnel compartment.

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3.4.19.5 Fire extinguishers. Fire extinguishers shall contain a minimum of 90 percent of rated full charge. All seals shall be intact. DA Form 253 shall be completed and attached to each extinguisher (see 6.3).

3.4.19.6 Basic issue items (BII). Unless otherwise specified (see 6.2), BII shall be processed, packaged and packed in accordance with MIL-B-12841. BII shall be stowed and secured in accordance with 3.4.19.6.1.

3.4.19.6.1 Stowage and securement of BII. BII and items removed for shipment shall be identified to the pertinent vehicle by serial number. BII shall be stored inside buildings, except during shipment. Packed BII and removed items shall be placed within the personnel compartment of vehicle. Large wooden boxes shall be placed on the vehicle floor and shall be secured with 1.25 inch wide strapping conforming to type I, class 1, finish A of QQ-S-781. Strapping shall be secured to holding devices within the compartment. Smaller wooden boxes shall be secured to the larger boxes with 0.75 inch wide strapping conforming to type I, class 1, finish A of QQ-S-781. All boxes shall be secured in such a manner as to prevent any movement during transit and to prevent damage to containers or vehicle interiors. Corner protectors shall be used under all strapping.

3.4.19.7 Access plate, gasket, and drain plugs. Access plate, gasket and preserved drain plugs (see 3.4.17.1) shall be packaged in a box conforming to type CF, Class Weather-Resistant of PPP-B-636. The box shall be closed with tape conforming to type III, class 1 of PPP-T-60, identified as to contents, and securely stowed within personnel compartment.

3.4.19.8 Packaging of tow hooks. Tow hooks and related hardware shall be removed for shipment and packaged in a type CF, Class Weather-Resistant box conforming to PPP-B-636. The box shall be closed with tape conforming to type III, class 1, of PPP-T-60, identified as to contents, and securely stowed within personnel compartment.

3.4.20 Vehicle closure.

3.4.20.1 Closure kit. Unless otherwise specified (see 6.2), each vehicle shall be provided with a vehicle protective closure kit. Closure kit shall be fabricated, assembled, and installed in accordance with the appendix.

3.4.20.1.1 Closure marking. The information "TO LIFT VEHICLE FOR LOADING, UNFASTEN AND ROLL COVER FROM FRONT AND REAR BOWS EXPOSING LIFTING EYES - AFTER LOADING, SECURE COVER IN THE ORIGINAL POSITION" shall be stenciled on teh front and rear of the closure in characters a minimum of 0.75 inch high using white enamel conforming to TT-E-529.

3.4.20.1.2 Closure disposition marking. The following information shall be stenciled on the outside, front and rear of the cover: "REUSABLE CLOSURE (COVER AND FRAMEWORK) - DO NOT DESTROY - WHEN REMOVED AND NO LONGER REQUIRED FOR PROTECTION, PACKAGE AND SHIP PER INSTRUCTIONS ON INSIDE OF COVER." The following information shall be stenciled on the inside, front and rear of the cover:

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PACKAGING AND SHIPPING INSTRUCTIONS

1. DEPOT BII CONTAINER MAY BE USED AS A SHIPPING BOX IF CLOSURE KIT WILL FIT WITHIN DIMENSIONS OF CONTAINER.
2. SECURELY BUNDLE LIKE ITEMS.
3. PACKAGE HARDWARE IN CLOTH BAGS.
4. GROUP LARGEST, HEAVIEST ITEMS IN BOTTOM OF BOX.
5. PLACE OTHER ITEMS IN VOID BETWEEN ABOVE ITEMS.
6. FOLD COVER - PLACE ON TOP OF ABOVE ITEMS.
7. IMMODILIZE ALL ITEMS AND SECURE COVER.
8. SHIP TO (address to be furnished by contracting officer).

All stenciled characters are to be a minimum of 0.75 inch high using white enamel conforming to TT-E-529.

3.4.20.1.3 Depot BII box marking. The information "DO NOT DESTROY - USE FOR RETURN SHIPMENT OF VEHICLE CLOSURE KIT" shall be stenciled on the depot BII box in lettering a minimum of 0.75 inch high with a contrasting color of enamel conforming to TT-E-529. Do not mark BII box if it is ascertained that the container will not accomodate the closure kit.

3.5 Level B. Vehicles shall be processed in the same manner as specified for Level A, with the following exceptions.

3.5.1 Transmission, transfer assembly, control differential, and final drives. The transmission, transfer assembly, control differential, and final drives shall contain operational lubricant as specified on applicable drawings, specifications, or lubrication order, filled to operating level. If, however, these units contain lubricating oil conforming to grades PE 10-1 or PE 30-1 of MIL-L-21260, an additional amount of the same oil shall be added to attain the operating level. Operating lubricants shall not be mixed with MIL-L-21260 except in an emergency. DD Form 1397 shall be annotated to indicate grade of lubricant or preservative oil used (see 3.4.1.1 and 3.4.8).

3.5.2 Engine crankcase. The engine crankcase shall contain normal operational lubricant as specified in lubrication order, filled to operational level. DD Form 1397 shall be annotated to indicate grade of lubricant used (see 3.4.1.1 and 3.4.10).

3.5.3 Engine preservation. The engine shall require no preservation for Level B shipment and storage (see 3.4.11).

3.5.4 Personnel heater and fuel pump. The personnel heater and fuel pump shall be in a ready to use condition. The heater exhaust opening shall be closed with a plastic plug conforming to MIL-C-5501/7 (see figure 8). A warning tag, bearing the information "HEATER EXHAUST OPENING CLOSED, REMOVE PLUG BEFORE OPERATING", shall be attached to the heater controls (see 3.4.12).

3.5.5 Residual fuel. Unless otherwise specified (see 6.2), vehicle shall be shipped without draining residual fuel from the fuel tank. The fuel tank shall not be preserved (see 3.4.13).

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3.5.6 Engine access compartment panels. Engine compartment panels in crew compartments shall not be removed (see 3.4.17.2).

3.5.7 Backrests and seats. Cushions of backrests and seats shall not be covered. If cushions are received packaged, they shall be stowed as received, in the crew compartment (see 3.4.19.3).

3.5.8 Tow hooks. Tow hooks shall be removed for overseas shipment only (see 3.4.19.8).

3.5.9 Vehicle closure. Vehicle closure kits shall not be provided for Level B shipment and storage (see 3.4.20).

3.5.10 Vision block openings. Plugs conforming to PPP-C-1752, type I, class 2, shall be installed in the vision block opening (see figure 8 and 8-1).

3.5.11 Bilge pump outlets. Bilge pump outlets shall be closed with plastic plugs conforming to MIL-C-5501/7 (see figure 8). A warning tag, bearing the information "BILGE PUMP OUTLETS CLOSED: PRIOR TO OPERATING BILGE PUMP, REMOVE PLUGS FROM OUTLETS", shall be secured to the bilge pump operating switch.

3.6 Loading. Loading of vehicles processed according to levels A and B shall be the same, except where differences are specified.

3.6.1 Loading flat cars. Loading of vehicles on open top railroad cars shall be in accordance with the applicable requirements of Section 1, General Rules Governing the Loading of Commodities on Open Top Cars, and figure 87 or 87A, Section 6, Rules Governing the Loading of Department of Defense Material on Open Top Cars, publication of the Association of American Railroads.

3.6.2 Highway shipment. Loading of vehicles for shipment by haulaway, and rules for shipment by driveaway or towaway shall be in accordance with Interstate Commerce Commission publication "Federal Motor Carrier Safety Regulations", and applicable military publications.

3.6.3 Drive-on/drive-off capability. When specified (see 6.2), the vehicle shall be capable of operation during loading and unloading. When the vehicle is to be operated for loading or unloading, the following provisions shall apply.

3.6.3.1 Additional fuel. Additional fuel shall be added, as required, to accomplish movement of the vehicle.

3.6.3.2 Batteries and electrolyte. Batteries shall be filled with electrolyte, fully charged, and battery cables connected. After vehicle self-movement for loading or placement in storage, the ground cable at the battery shall be disconnected and then secured to the battery carrier with 0.75 inch tape conforming to type IV of PPP-T-97. A tag bearing the following message, "VEHICLE PRESERVED FOR DRIVE-AWAY CONDITION. BEFORE CRANKING, CONNECT GROUND CABLE TO BATTERY TERMINAL. ENGINE AND FUEL TANKS NOT PRESERVED.", shall be located in a conspicuous location in the driver's compartment.

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3.6.4 Reprocessing engine after loading.

3.6.4.1 Level A. If the engine is operated in connection with the moving of vehicle to loading area or during the loading itself, the engine shall be reprocessed as specified in 3.4.11.1 through 3.4.11.6. The vehicle cover shall be rolled clear of the engine intake and exhaust to provide air circulation and to prevent damage to the cover. After reprocessing of the engine, the vehicle cover shall be replaced in its original position.

3.6.4.2 Level B. If the engine is operated in connection with movement of the vehicle for loading or unloading, there shall be no additional processing of the engine.

3.7 Marking. In addition to any special marking required in the contract (see 6.2), the vehicle shall be marked in accordance with MIL-STD-129.

3.7.1 Lifting points. The legend "LIFT HERE" with arrow pointing to the lifting eye shall be stenciled adjacent to each lifting eye using black ink conforming to TT-I-1795.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection records. The contractor shall maintain records of all inspections performed and such records shall be readily available for review by the Government representative.

4.2 First article inspection. One of the first ten production processed vehicles (see 3.1) shall be subjected to the inspections specified in 4.5.1 through 4.5.5 to determine conformance to the requirements of table I.

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TABLE I. Processing requirements.
(See indicated paragraphs for Level A, B requirements)

Component	Cleaning	Preservation		Packaging/Stowage	
	Levels A & B	Level A	Level B	Level A	Level B
Record forms				3.4.1.1	3.4.1.1
Processing records				3.4.1.2	3.4.1.2
Disassembly				3.4.4	3.4.4
Matchmarking				3.4.4.1	3.4.4.1
Interior of vehicle	3.4.5.1				
Battery supports and retainers	3.4.5.1.1	3.4.7	3.4.7		
Backrests and seats	3.4.5.1.2			3.4.19.3	3.5.7
Vehicle exterior	3.4.5.2				
Relubrication		3.4.6	3.4.6		
Transmission, transfer assembly, control differential and final drives 1/		3.4.8	3.5.1		
Cooling system 1/		3.4.9	3.4.9		
Water and antifreeze procedure		3.4.9.1	3.4.9.1		
Water and corrosion inhibitor procedure		3.4.9.2	3.4.9.2		
Water compound procedure		3.4.9.3	3.4.9.3		
Engine crankcase 1/		3.4.10	3.5.2		
Compression ignition engine		3.4.11	3.5.3		
Preservation through fuel system and combustion chamber		3.4.11.1	3.5.3		
Engine purging		3.4.11.2	3.5.3		
Preservation through air intake and exhaust system 1/		3.4.11.3	3.5.3	3.4.11.3	
Preservation through oil level gage rod opening		3.4.11.4	3.5.3	3.4.11.4	
Preservation through flywheel housing		3.4.11.5			
Air cleaner				3.4.11.6	
Personnel and engine compartment heater and lines		3.4.12	3.5.4	3.4.12	3.5.4
Fuel tank		3.4.13	3.5.5		
Ramp winch assembly		3.4.14	3.4.14		

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TABLE I. Processing requirements - Continued.
(See indicated paragraphs for Level A, B requirements)

Component	Cleaning	Preservation		Packaging/Stowage	
	Levels A & B	Level A	Level B	Level A	Level B
Ramp hydraulic reservoir		3.4.14.1	3.4.14.1		
Turret opening				3.4.15	3.4.15
Hatches and doors		3.4.16	3.4.16	3.4.16	3.4.16
Engine compartment access plate		3.4.17.1	3.4.17.1	3.4.19.7	3.4.19.7
Engine compartment access panels				3.4.17.2	3.5.6
Miscellaneous preservation		3.4.18	3.4.18		
Dry charged batteries and cables		3.4.19.1	3.4.19.1	3.4.19.1	3.4.19.1
Electrolyte				3.4.19.2	3.4.19.2
Periscopes				3.4.19.4	3.4.19.4
Fire extinguishers				3.4.19.5	3.4.19.5
Basic issue items				3.4.19.6	3.4.19.6
Strapping				3.4.19.6.1	3.4.19.6.1
Tow hooks				3.4.19.8	3.5.8
Vision block openings					3.5.10
Bilge pump outlets					3.5.11
Vehicle closure kit				3.4.20.1	3.5.7
Closure marking				3.4.20.1.1	
Depot BII box marking				3.4.20.1.3	
Loading on flat cars				3.6.1	3.6.1
Highway shipment				3.6.2	3.6.2
Reprocessing engine after loading				3.6.4.1	3.6.4.2
Marking				3.7	3.7
Lifting points				3.7.1	3.7.1

1/ Inspect DD Form 1397

4.2.1 Failure. Failure of any first article samples to conform to the applicable requirements of this specification shall be cause for rejection of the vehicles by the Government. The vehicle shall not be accepted until objective evidence that the contractor has corrected the condition causing the failure has been provided to the Government.

4.3 Quality conformance tests.

4.3.1 Acceptance tests. Unless otherwise specified (see 6.2), one processed vehicle selected at random from each day's production shall be tested in accordance with 4.5.2, 4.5.3 and 4.5.5.

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4.3.2 Control tests. Control tests shall be conducted on one processed vehicle from each lot of 50 units consecutively processed, except that not more than one test in a six month period. The processed vehicle shall be subjected to the inspections specified in 4.5.1 through 4.5.5.

4.4 Failure. Failure of any processed vehicle to pass any of the specified tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.5 Methods of inspection.

4.5.1 Materials. To determine conformance to 3.2, all materials to be used in processing of vehicles shall be inspected in accordance with the material specifications. Certified inspection and laboratory test reports shall be provided which show that furnished materials conform to the applicable material specifications, except for materials that have been inspected by the Government at the source. Materials listed on a Qualified Products List shall be obtained from one of the approved sources indicated.

4.5.2 Cleaning. To determine conformance to 3.4.5.1, the vehicle interior shall be examined for cleanliness in accordance with the applicable provisions of MIL-P-116. To determine conformance to 3.4.5.2, the vehicle exterior shall be examined for cleanliness. Surfaces on which tape is to be applied shall be examined for cleanliness before application.

4.5.3 Cooling system. To determine conformance to 3.4.9, the engine coolant shall be tested using a hydrometer-thermometer type tester, with a range of minus 60°F to plus 160°F, conforming to GG-T-250.

4.5.4 Engine. To determine conformance to 3.4.11.1 through 3.4.11.3, as applicable, the interior of the engine shall be examined for surface coverage. The engine shall be disassembled to the extent necessary to permit visual examination of all surfaces within the combustion chamber. The combustion chamber is all surfaces within the cylinder from and including the surface of the head within the cylinder. All surfaces within the combustion chamber shall have a wet coating of preservative oil as obtained when the item is dipped or flushed with the oil. The processing method of the approved preserved engine shall be submitted in writing for use in processing engines in all subsequent production vehicles.

4.5.5 Fuel tank. To determine conformance to 3.4.13, visual inspection of preservative application shall be accomplished to assure addition of the correct amount of oil, based upon a known amount of residual fuel.

5. PACKAGING

This section is not applicable to this specification.

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6. NOTES

6.1 Intended use. This specification covers processing of the M741 and M741A1 vehicle for storage outside of buildings, for immediate shipment and domestic or overseas shipment, including carloading (see 1.1).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Level of processing required (see 1.2).
c. If first article samples are not required (see 3.1).
d. Required procedure for protection of cooling system (see 3.4.9).
e. If BII should be processed, packaged, packed and stored other than as specified (see 3.4.19.6).
f. If vehicle closure kit is not required (see 3.4.20.1).
g. If draining of residual fuel from fuel tank is required (see 3.5.5).
h. If drive-on/drive-off capability is required (see 3.6.3).
i. If special marking is required (see 3.7).
j. If responsibility for inspection shall be other than as specified (see 4.1).
k. If inspections other than those specified are required (see 4.3.1).

6.3 Safety precautions. Caution should be exercised in handling carbon dioxide (CO_2) fire extinguisher cylinders. Cylinders should not be dropped, permitted to strike each other, or handled roughly. Extreme care should be exercised during the reinstallation operation to avoid tripping fire extinguisher control trigger (see 3.4.19.5).

6.4 Forms. A copy of the "Equipment Log Book" and all required forms i.e., DD Form 1397 (see 3.4.1.1) and DA Form 253 (see 3.4.19.5) will be furnished to the contractor by the Government at least 30 days before shipment of the vehicles required by the contract delivery schedule.

6.5 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

6.6 Dimensions. Unless otherwise specified all dimensions are in inches.

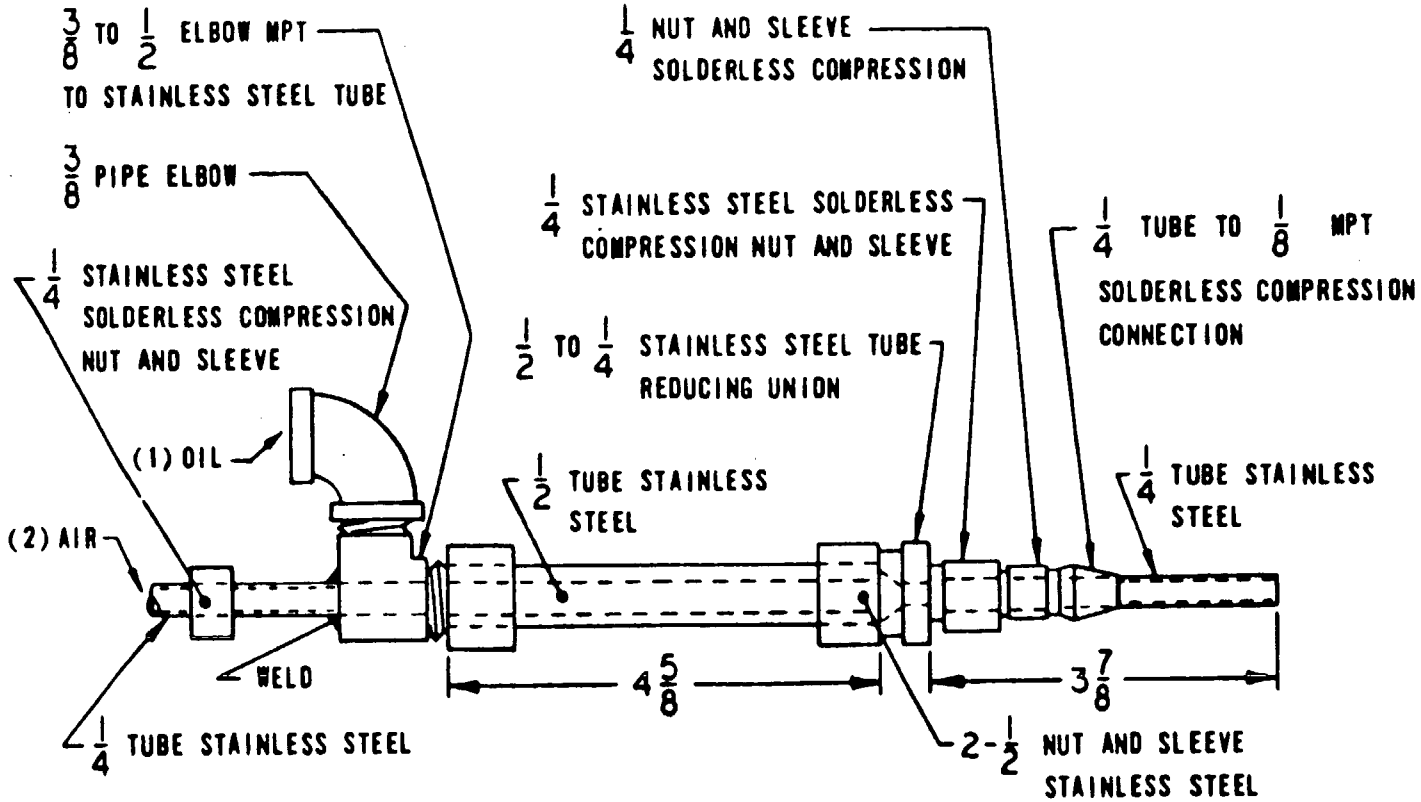
6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AT

Preparing activity:
Army - AT

Project No. PACK-A311

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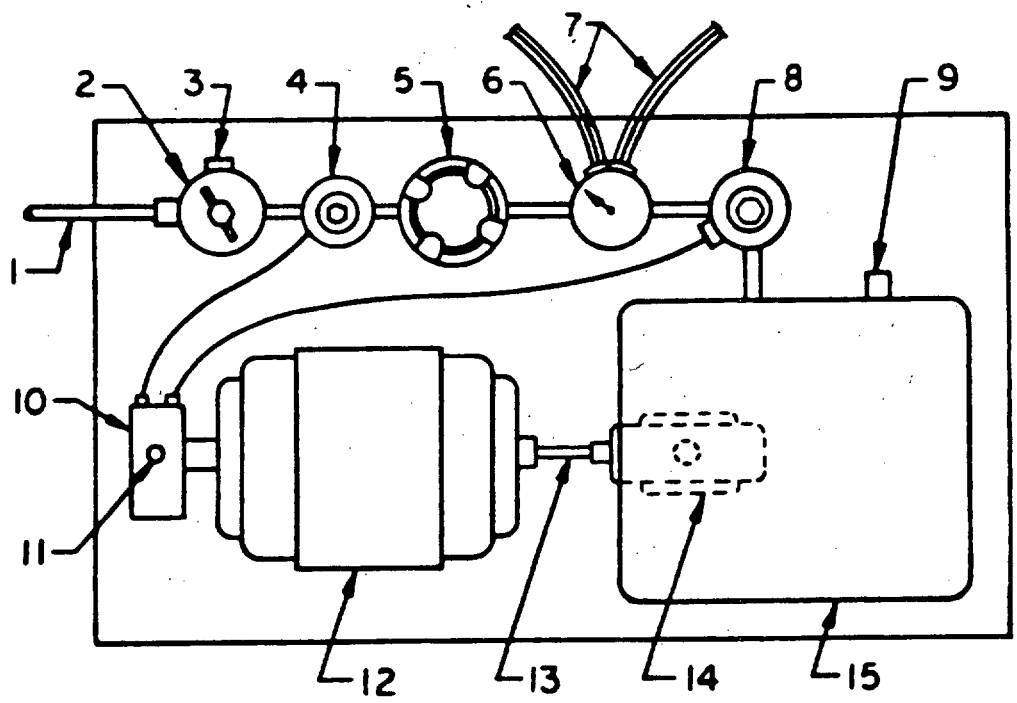


NOTE: (1) AND (2) CONNECT TO CORRESPONDING LINES ON FIGURE 3

FABRICATION DETAILS FOR OIL SPRAY ATOMIZING NOZZLE

FIGURE 1. Atomizing device.

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1. AIR LINE
2. AIR PRESSURE REGULATOR
3. AIR PRESSURE GAGE
4. SOLENOID VALVE
5. MOISTURE SEPARATOR
6. OIL PRESSURE GAGE
7. TWO DOUBLE TAPED LINES
8. SOLENOID VALVE
9. OIL TANK LEVEL GAGE
10. ELECTRIC JUNCTION BOX
11. ELECTRIC LINE
12. MOTOR 1/4 H P
13. SHAFT
14. POSITIVE DISPLACEMENT OIL PUMP
15. OIL TANK

NOTE:
THIS EQUIPMENT HAS PROVEN SATISFACTORY FOR ATOMIZE SPRAYING OF PRESERVATIVE OIL IN CONJUNCTION WITH FIGURE 2

FIGURE 2. Pressure pump

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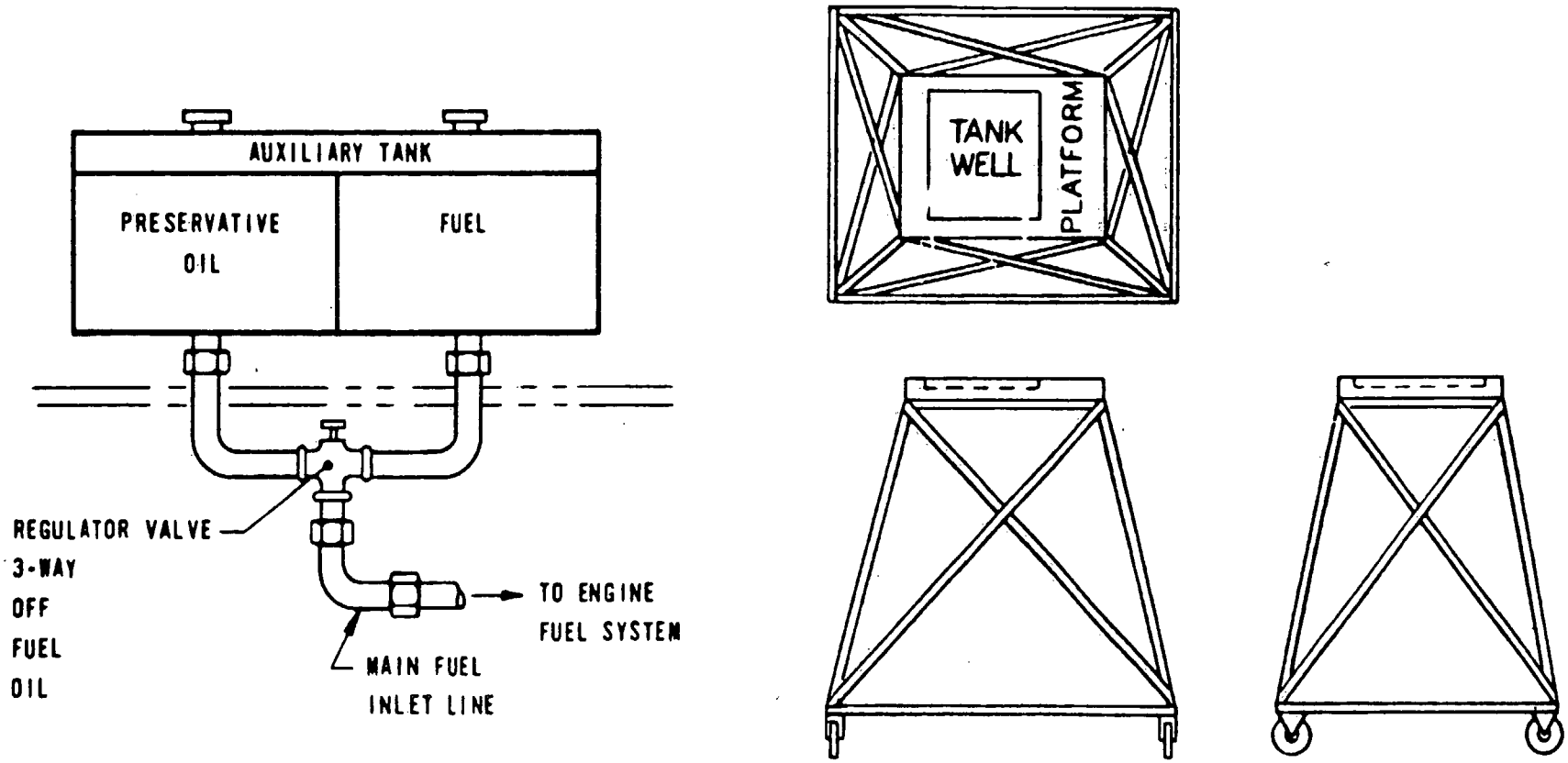
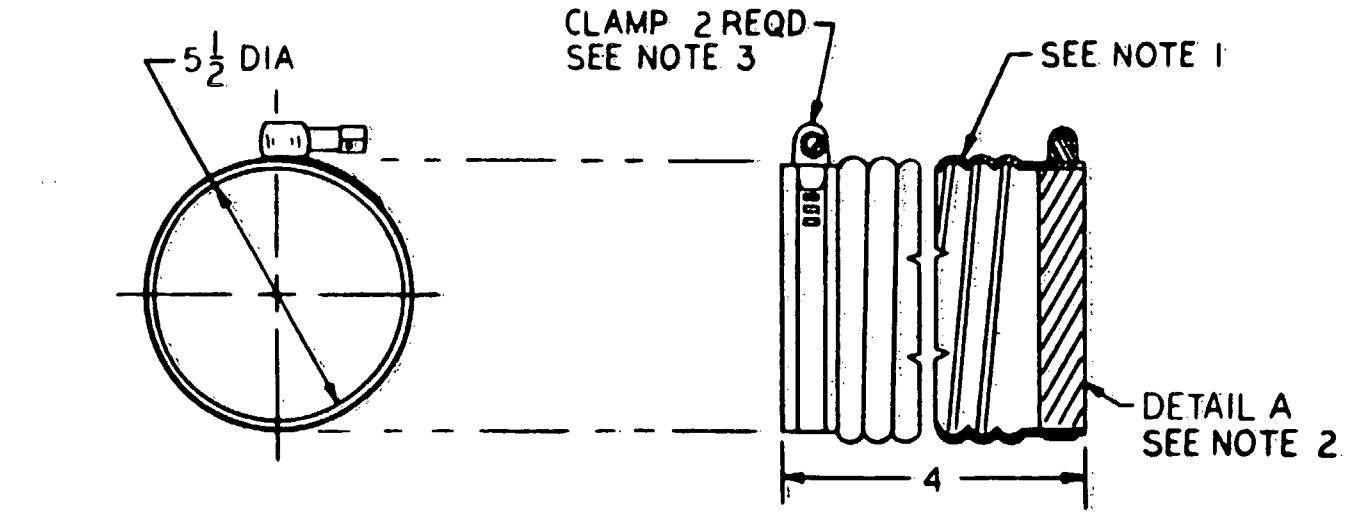
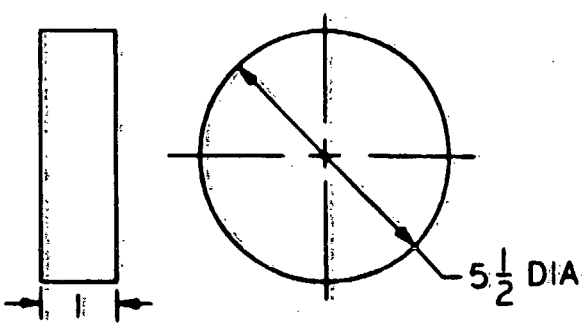


FIGURE 3. Auxiliary fuel container equipment.

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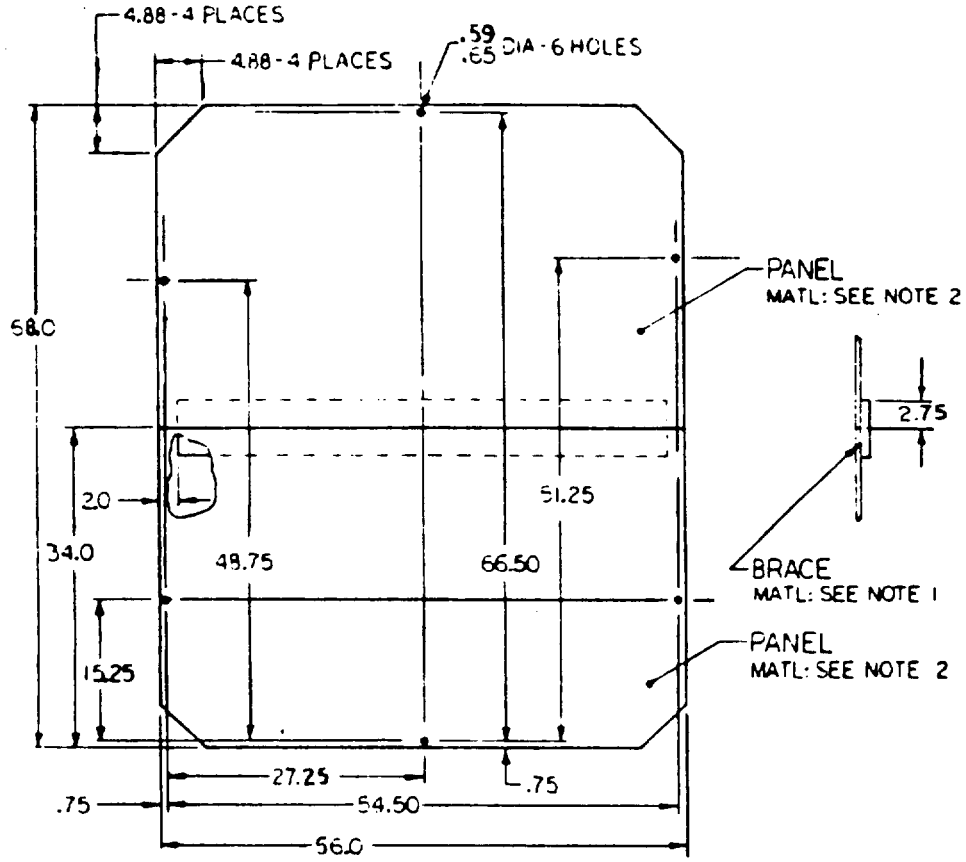
DETAIL A-1 REQD

NOTES:

1. HOSE FLEXIBLE TO BE MADE FROM MATERIAL CONFORMING TO MIL-H-52079 OR EQUIVALENT.
2. PLUG TO BE MADE FROM 1 INCH PLYWOOD, NN P 530, STD. INT. WITH EXT. GLUE.
3. USE MS35842-16 OR EQUIVALENT CLAMPS.

FIGURE 4. Boot air restrictor.

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NOTES:

1. MATERIAL:
   WOOD MEMBER, GROUP I OR II
   MIL-STD-731, 1x6 NOMINAL,
   52.50-51.50 LONG

2. MATERIAL:
   PLYWOOD, INTERIOR GRADE.
   EXTERIOR GLUE, SPEC. NN-P-530,
   0.750 THK

3. TOLERANCE .XX±0.06, x±.01
   DIMENSIONS IN INCHES

4. ATTACH PANELS TO BRACE WITH
   5d NAILS, TYPE II, STYLE 4, SPEC.
   FF-N-105, CLINCHED, SPACED
   3.00, STAGGERED EACH SIDE OF
   SPLICE

FIGURE 5. Turret cover

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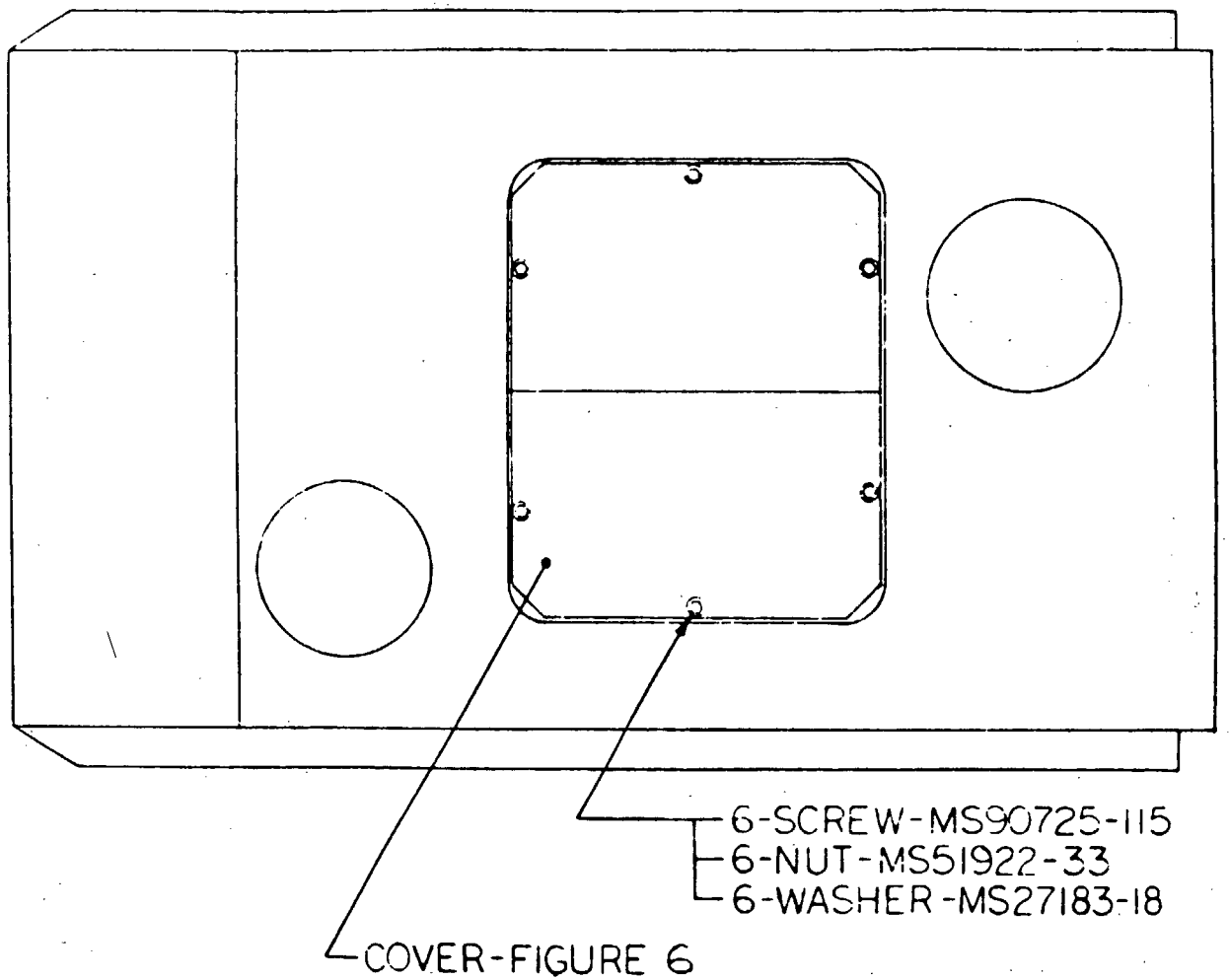


FIGURE 6. Turret cover installation.

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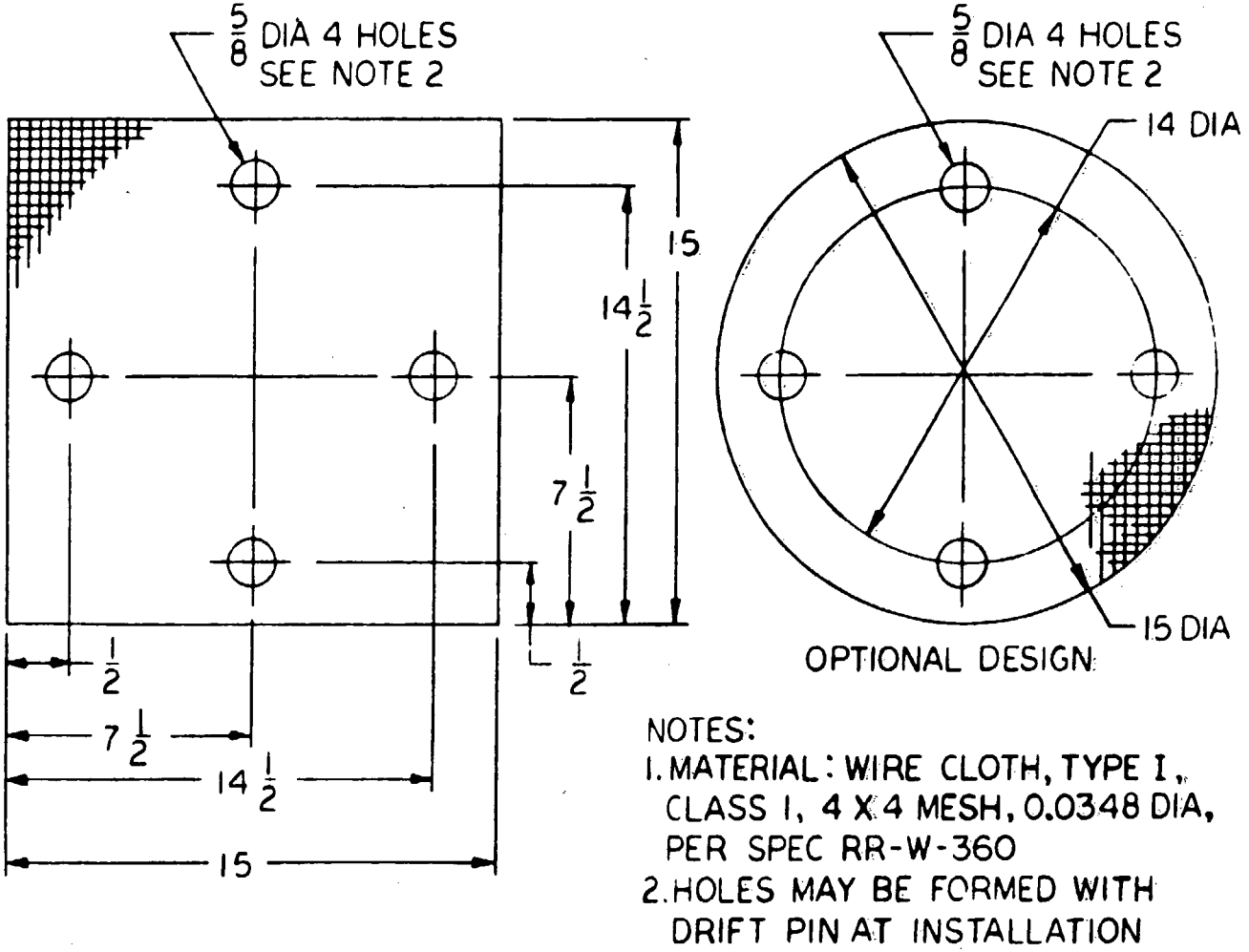
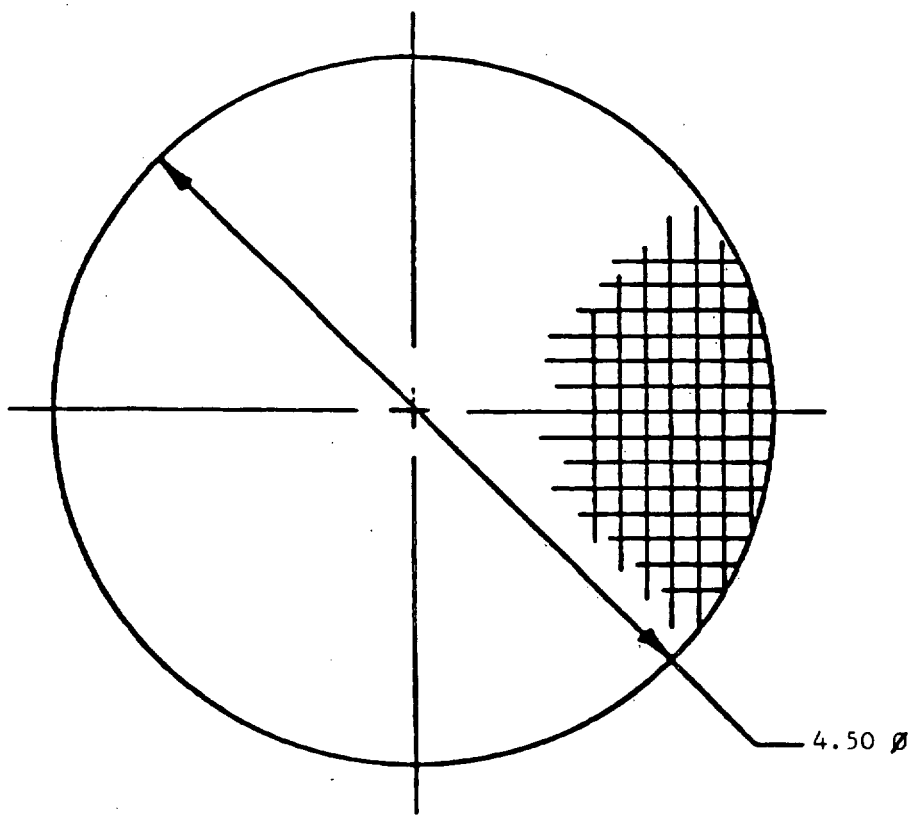


FIGURE 7. Screen.

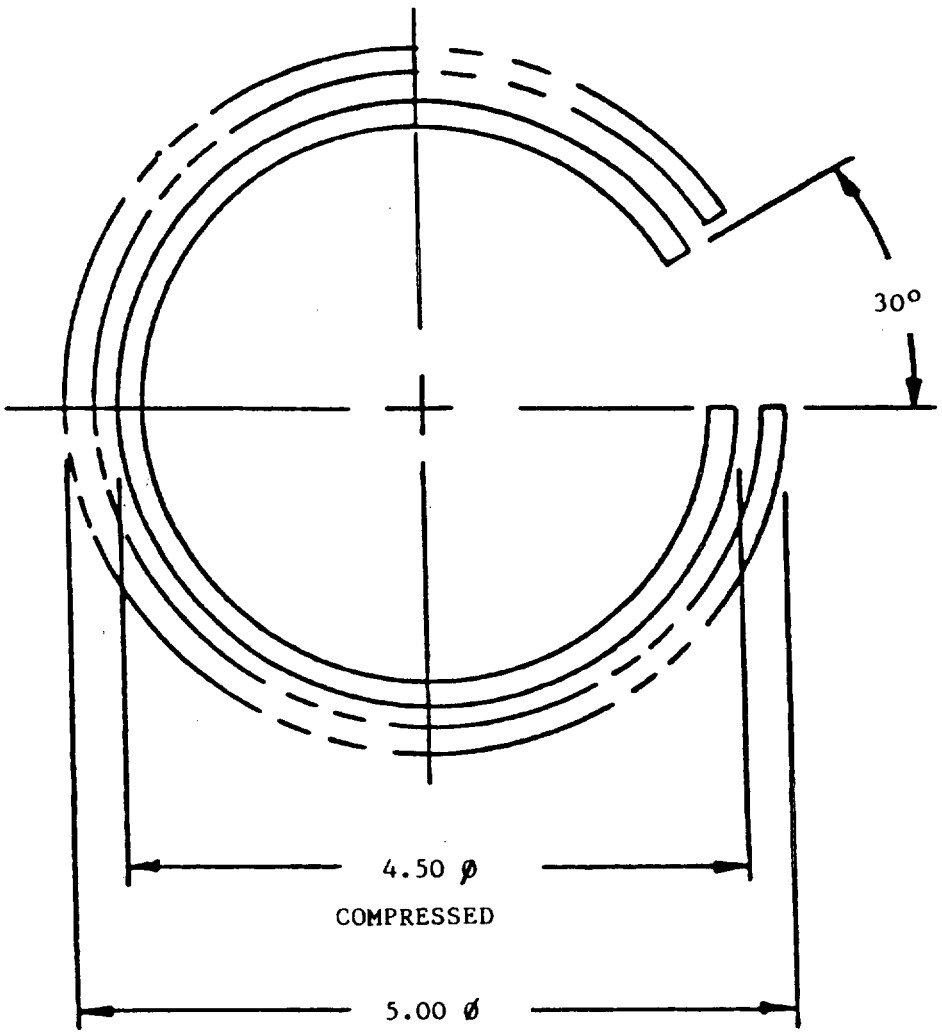
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NOTE:
1. MATERIAL: WIRECLOTH, TYPE I, CLASS 1, 4x4 MESH, 0.0348 DIA, SPEC. RR-W-360

FIGURE 7A. Screen.

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NOTE:
1. MATERIAL: WIRE, STEEL, MUSIC SPEC. QQ-W-470. 0.156 DIA.

FIGURE 7B. Spring retainer.

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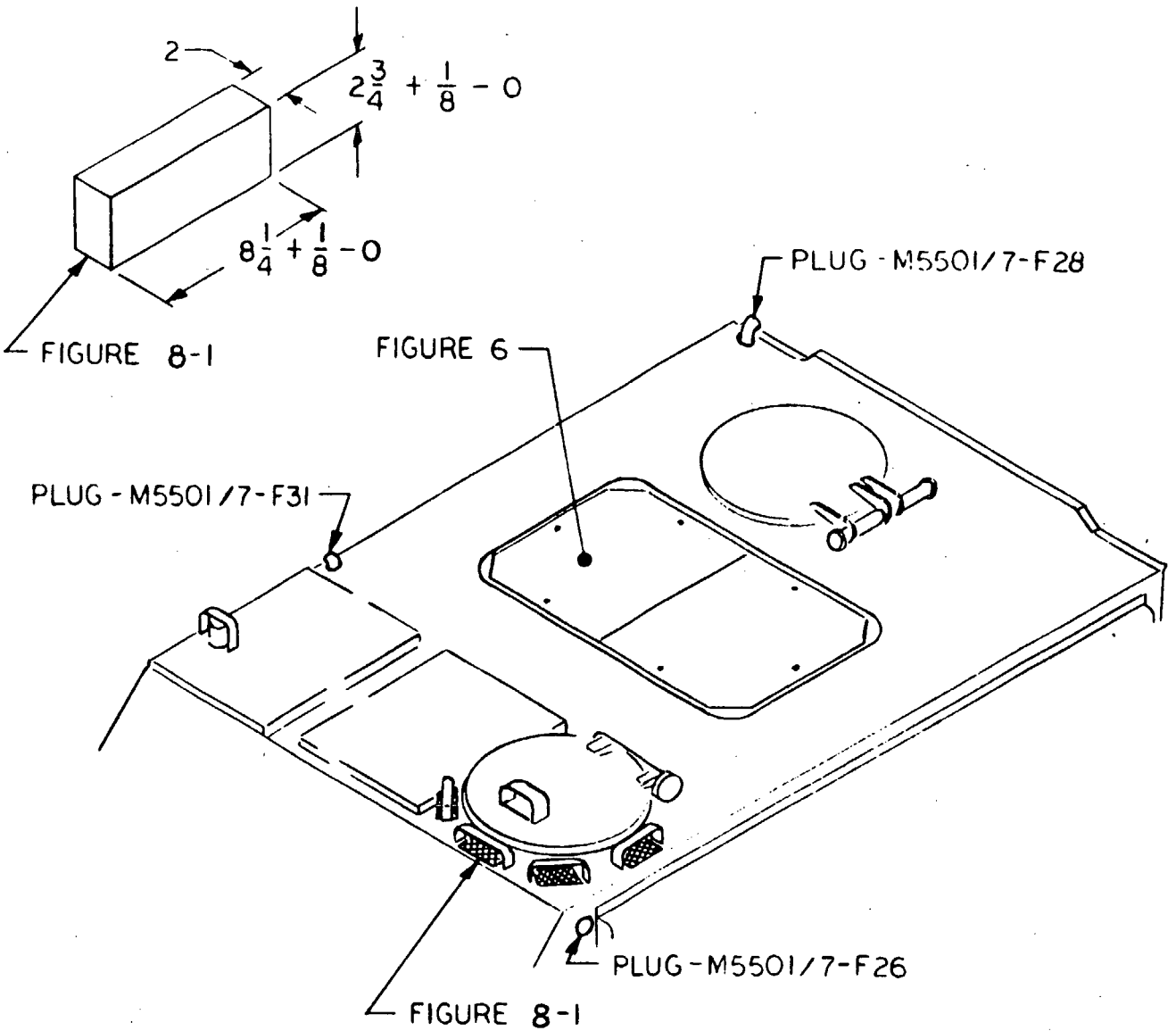


FIGURE 8. Level B closure

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APPENDIX

REQUIREMENTS FOR FABRICATING, ASSEMBLING, INSTALLING, REMOVING, DISASSEMBLING, AND PRESERVING, PACKAGING, AND PACKING VEHICLE CLOSURE FOR M741 AND M741A1 VEHICLE

10. SCOPE

10.1 Scope. This appendix covers the requirements for fabrication, assembling, installing, removing, and disassembling the vehicle closure and for preserving, packaging, and packing the disassembled vehicle closure for the M741 and M741A1. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance only.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

MS 21044	- Nut, Self-Locking, Hexagon-Regular Height, 250 degrees Fahrenheit (°F), 125 Ksi Ftu and 60 Ksi Ftu.
MS 21083	- Nut, Self-Locking, Hexagon, Non-Metallic Insert, Low Height, 250 degrees Fahrenheit (°F).
MS 27183	- Washer, Flat-Round, Steel, Cadmium Plated, General Purpose (in/mm).
MS 35207	- Screw, Machine-Pan Head, Cross-Recessed, Carbon Steel, Cadmium Plated, UNF-2A (in/mm).
MS 51922	- Nut, Self-Locking, Hexagon-Prevailing Torque, General Purpose, 250 degrees Fahrenheit (°F), UNC-2B and UNF-2B.
MS 90725	- Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Grade 5, Cadmium Plated, UNC-2A.
MS 90726	- Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Grade 5, Cadmium Plated, UNF-2A.

20.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

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APPENDIX

DRAWINGS
ARMY

11678121 - Closure Kit, Vehicle Protective.

PACKAGING DATA SHEET

11678121 - Closure Kit, Vehicle Protective.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity, or as directed by the contracting officer.)

30. REQUIREMENTS

30.1 Closure kit. The vehicle closure kit No. 11678121 consists of a frame assembly No. 10917814 and a cover No. 10917812. After the frame assembly has been installed on the vehicle, the cover is attached to the frame. The drawing numbers specified in the succeeding paragraphs are referenced on these assembly drawings.

30.2 Fabrication of frame. The frame assembly components, when permitted by optional material callouts on the drawings, may be either steel or aluminum. Steel parts in direct contact with aluminum must be painted to avoid electrolysis.

30.3 Attitude of fasteners. Except as otherwise specified in 30.4 or in the drawings, all fasteners in the frame assembly shall be installed with the threaded end pointing down or toward the center of the assembly.

30.4 Cover. The cover shall be fabricated in accordance with Drawing 10917812.

30.4 Assembly of frame. The frame may be assembled using appropriate jigs and fitures. This is recommended especially for large production runs. If a jig or fixture is not used, the frame may be assembled as follows:

a. Form a subassembly by joining one front frame section 10917732-1 or 10917732-3 and one front frame section 10917732-2 or 10917732-4, using connector 10917850, 6 screws MS 90725-8 and 6 nuts MS 51922-1.

b. Form a subassembly by joining one front right frame section 10917392 or 10917392-1 and one rear frame section 10917394 or 10917394-1 using connector 10917850, 6 screws MS 90725-8 and 6 nuts MS 51922-1. Lower 2 screws shall be positioned with threaded ends pointing out.

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c. Form a subassembly by joining one rear frame section 10917828-1 or 10917828-3 and one rear frame section 10917828-2 or 10917828-4, using connector 10917850, 6 screws MS 90725-8 and 6 nuts MS 51922-1.

d. Form a subassembly by joining one front left frame section 10917391 or 10917391 and one rear left frame section 10917393 or 10917393-1 using connector 10917850, 6 screws MS 90725-8 and 6 nuts MS 51922-1. Lower 2 screws shall be positioned with threaded ends pointing out.

e. Join together at the corners the four subassemblies formed in steps (a) through (d), using 4 connectors 10917706, 4 screws MS 90726-68 and 4 nuts MS 21044-6N6. Screws shall be positioned with the threaded ends pointing up.

f. Attach 4 brackets 10917849 to the inside of the side frame members using 4 connectors 10917706, 4 screws MS 90726-68 and 4 nuts MS 21044-6N6. Screws shall be positioned with the threaded ends pointing up.

g. Attach remaining 4 connectors 10917706 to the top side of the side frame members between the brackets installed in step (f) using 4 screws MS 90726-68 and 4 nuts MS 21044-6N6. Screws shall be positioned with the threaded ends pointing up.

h. Attach 4 plates 10917396 to the outside of side frame members using 16 screws MS 90725-8 and 16 nuts MS 51922-1. The screws at location of section J-J shall be positioned with the threaded ends pointed out.

i. Attach 20 fasteners 10922144 to the outside of the frame members using 20 screws MS 35207-65 and 20 nuts MS 21083-N3.

j. Form a ridge pole by joining the two top tubing sections 10917731 or 10917729 using connector 10917707-1 or 10917707-2, 2 screws MS 90725-12 and 2 nuts MS 51922-1.

k. Attach 12 tubes 10917737-1 or 10917737-2 to the ridge pole formed in step (j), using 12 screws MS 90725-12 and 12 nuts MS 51922-1.

l. Lift the completed bow section formed in step (k) and slide the ends of the tubes over the connectors 10917706 on the frame assembly. See sections C-C and D-D, Drawing 10917814. This completes the frame assembly. For ease of handling, this step may be accomplished after installing frame on vehicle per 30.6.

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30.6 Installation of frame assembly on vehicle.

a. Lower the trim vane, raise and lock the ramp, and close all hatches and cupolas.

b. Lift the frame assembly and place on vehicle top so that slots in the rear frame members fit around the ramp skids.

c. Loosely attach 4 supports 11678119 to the four plates 10917396 on the frame assembly using 4 screws MS 90725-62, 8 washers MS 27183-14 amd 4 nuts MS 51922-17.

d. Attach the lower ends of the supports installed in step (c) to the flotation pod by removing the 4 pod bolts and washers that line up with the holes in the supports, placing 3 washers MS 27183-18 between the pod and supports and reinstalling the removed bolts and washers. Tighten the top support fasteners installed in step (c).

e. If the bow assembly was not installed per step 30.5 (1), it shall now be installed.

30.7 Installation of cover.

a. Unfold the cover 10917812 and place over the bow assembly with the vent covers to the outside.

b. Fold up corner flaps on the cover and insert 10 rods 10890663 into flaps through the slots. Clamp rods in place with the 20 fasteners 10922144. Fold down corner flaps.

c. Raise trim vane.

d. This completes the closure kit installation.

30.8 Removal and disassembly. The closure kit shall be removed from the vehicle and disassembled in accordance with 30.8.1 and 30.8.2.

30.8.1 Cover.

a. Lower trim vane.

b. Fold up cover flaps and release the 20 rod clamps.

c. Withdraw the 10 rods from the cover.

d. Remove cover from frame and bows.

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30.8.2 Frame and bow assembly.

a. Lift bow assembly from frame.

b. Remove the 4 screws and 16 washers which attach the 4 frame supports to the flotation pod to disconnect the supports. Reinstall 4 screws and 4 washers in the flotation pod.

c. Remove 4 screws and nuts and 8 washers to disconnect frame supports from frame.

d. Lift frame from vehicle.

e. Raise trim vane.

f. Disassemble frame and bow assembly by removing screws and nuts from all joints.

30.9 Preserving, packaging, and packing. The disassembled vehicle closure shall be preserved, packaged, and packed in accordance with Packaging Data Sheet 11678121.

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